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RESEARCH ARTICLE

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Do Pay-for-Grades Programs Encourage Student Academic Cheating? Evidence from a Randomized Experiment

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Abstract Pay-for-grades programs aim to incentivize families and students to engage in specified behaviors given financial remittance. In addition to doubts over its effectiveness, educators worry that monetary incentives could skew student learning motivation and lead to academic cheating. Using a randomized control trial in 11 Chinese primary schools, we studied the effects of pay-for-grades programs on academic cheating. We randomly assigned 82 classrooms into treatment or control conditions, and used a statistical algorithm to determine the occurrence of cheating. While our data indicates a concerning level of cheating overall, our experiment did not find any relationship between academic cheating and the pay-for-grades program. Finally, we discuss the feasibility of such policies in improving learning outcomes and its policy implications.

Keywords pay-for-grades, cheating, randomized control trial, China, primary school

Introduction

Using financial incentives to stimulate student educational outcomes is not a new idea. Ravitch (2000) has documented the New York City school system's history

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of promising financial rewards to children on the basis of their school performance as early as the 1820s. Long regarded as a fringe idea in the education community, this approach has resurfaced as a developmental and education policy tool in recent years. There are many iterations of this approach (Slavin, 2010): Some schools use cash to boost school attendance, some use financial incentives to encourage children to maintain better grades, some for promoting educational attainment, while others use cash incentives to improve interaction among peers.

These policies hinge on the assumption that families and children will be incentivized to engage in specified behaviors given financial remittance (Bettinger, 2012). Researchers have debated the merits of such programs, but judging by current roll-outs, similar policies are likely to stay. According to an extensive review by Slavin (2010), more than a dozen various programs of similar nature are being tested in developing countries and among developed nations: Israel, the UK, and the USA are conducting experimentations on this approach.

But doubts over whether monetary incentives would skew student learning motivation and lead to unintended consequences, such as academic cheating, remains a concern. Specifically, when targeted behaviors are aspirational, such as improving grades at the end of a semester, it is possible that some students will cheat on exams rather than take the time to authentically learn the material (Cizek, 1999). In other cases, when the short-term stakes are high, invigilators, teachers, and administrators may even help students cheat (Jacob & Levitt, 2003). Though these are valid concerns in their own right, empirical evidence on this very issue is limited.

In this study, we evaluate the effects of a common cash transfer program, known as “pay-for-grades,” on academic cheating. Such programs are designed for the sole purpose of improving learning outcomes through paying students for good grades or test scores (Le, 2015). Using a randomized control trial in Chinese migrant primary schools in Beijing, we studied whether students who sit next to each other would copy from one another during standardized tests once they were incentivized by the pay-for-grades program. We studied academic cheating from a quantitative perspective. This appears to be a new line of research in China. We provide empirical evidence into the ability of this policy tool to improve learning outcomes. We concur that it is possible to have some

cheating going on in our chosen sites but the academic cheating is not strong enough to affect test scores. Lastly, we discuss its policy implications.

Literature Review

Pay-for-Grades in Schools

Traditionally, the focus of many education systems, including those in China, has been to improve school enrollment and keep children from leaving the system. This focus is reflected in the United Nation's millennium development goals (MDGs), where universal primary education is one of the priorities, and where the popular conditional cash transfer programs (CCTs) emerged as policy leverage. Building on the promise of expectancy value theory of achievement (Atkinson, 1957; Eccles, 1983; Wigfield, 1994), it was argued that directly paying parents and students would improve their motivation to academically invest in their education (See Morais de Sa e Silva, 2015 for a review). It did not take long, however, for many educational researchers to realize that the basic CCTs did not necessarily improve learning outcomes (e.g., Simões & Sabates, 2014). A few upgraded CCTs implemented in Nicaragua, the USA (Opportunity NYC), Bangladesh, Cambodia, and Pakistan went one step further and made CCTs contingent upon satisfactory student academic performance (Morais de Sa e Silva, 2015).

In addition to keeping students inside schools, there is also an increasing number of special CCTs designed with the goal of improving learning outcomes through paying students for good grades or test scores (so-called pay-for-grades programs). Scholarly interest in such programs has surged. In a meta-analysis, Le (2015) identified 18 randomized control trials (i.e., experiments) of pay-for-grades programs targeting disadvantaged students in the past decade. She found overall weak and positive achievement gains, but the gains varied depending on the subject matter being tested, with mathematics showing the most gains, while reading and language arts showed zero gains.

Despite these potential benefits, using money to motivate student learning may produce a negative side effect—cheating. Behrman, Parker, Todd, and Wolpin (2015) reported extensive student cheating when monetary rewards were promised if their mathematics scores were improved during a test. This side

effect is likely to be serious in developing countries whose educational systems are historically plagued by various forms of cheating and corruption (Alutu & Aluede, 2006; Heyneman, 2004; Heyneman, Anderson, & Nuraliyeva, 2007; Kobiowu & Alao, 2009; Osipian, 2009; Vlaardingerbroek, Shehab, & Alameh, 2011). Indian parents scaling the school wall to help their children cheat on an important secondary school exit exam in 2015 is just one recent example of the extreme families will endure in order to secure financial incentives (Dearden, 2015). However, despite discussion in Behrman et al. (2015), there has not been research on the relationship between cheating and student performance incentive programs.

Academic Cheating

Academic cheating is regarded to be a significant issue at all levels of education. Cizek (1999) reported that one-third of students in elementary schools admitted to cheating in some form. In the US, over 60 % of middle school and high school students believed that cheating was a serious issue in their school (Evans & Craig, 1990). Large-scale cheating by both teachers and students was repeatedly reported by the media and researchers (Davis, Drinan, & Gallant, 2009).

From an individual perspective, McCabe (2005) found that getting good grades is the chief motive for students. Researchers have consequently found that the occurrence of cheating is associated with a student's self-efficacy and how much they identify with the academic culture of a school (Finn & Frone, 2004; Murdock, Hale, & Weber, 2001). There were multiple reports cited in Cizek's review (1999) that cheating is inversely correlated with student achievement, though a recent study depicts a more complicated relationship (Finn & Frone, 2004).

Cheating also appears to be influenced by cultural factors. Previous studies suggest that cheating is more likely to be perceived as acceptable in more collective Asian societies like Japan (Diekhoff, Labeff, Shinohara, & Yasukawa, 1999) and the Middle East (McCabe, Feghali, & Abdallah, 2008; Vlaardingerbroek et al., 2011). China appears to fall in this same category as well. Based on case studies of essay writing behaviors of Chinese students attending foreign higher institutions, Gu and Brooks (2008) and Shei (2005) argue that in the Chinese learning culture, a substantial period of imitation is generally

emphasized for beginners. Cheating, taking the form of copying from one another, is thus seen more as an effort to achieve competency than a type of negative academic behavior.

From a sociological perspective, cheating could also be conceived as a means of rebellion toward perceived social injustice or alienation. Murdock et al. (2001) found that middle schoolers who perceived their teachers to be uncaring or incompetent were more likely to report cheating as a means to show that they could also break school rules. Based on his extensive fieldwork in Chinese migrant schools, as well as a considerable body of surveys and interviews, Xiong (2015) demonstrated the existence of an anti-school subculture in migrant schools where students entirely rejected the values of formal schooling. The students, who were denied admission into public school systems due to their migrant status, defied traditional learning culture. The normative environment of migrant students appears to be highly conducive to cheating.

To summarize, previous literature on academic cheating suggests motivational and sociological links to cheating behavior. Although pay-for-grades programs use financial incentives as the main motive, currently no research has empirically examined whether it increases academic cheating in schools or not. The migrant school setting in China provides an ideal testing-ground for this study, as students mostly come from the low socioeconomic backgrounds that are the target of pay-for-grades policies.

Research Methods

Experiment Design and Implementation

We designed this study based on the pay-for-grades experiment by Li, Han, Zhang, and Rozelle (2014), which was implemented in 11 Beijing migrant schools that were chosen randomly from a comprehensive list of 340 migrant schools in Beijing in August 2009. The details are omitted here for space concerns. For the original design including the rationale behind choosing these schools and backgrounds of these schools, readers are encouraged to read the first study.

The experimenters implemented two multiple-choice tests, the baseline test was given in September 2009, and the evaluation test in January 2010. In

addition, a survey was implemented at the baseline stage to collect basic background information. The experimenters also collected classroom seating tables.

The baseline test had approximately 30 math and 30 Chinese multiple-choice questions with each question having 2–4 choices. Students were given 45 minutes to answer the questions. All classes in the same grade used the same test, and different grades used different tests. All tests were designed by an external expert panel, and the tests were designed to resemble standard tests that students in Chinese schools were accustomed to taking. Students took the test during their regular class time and in their regular classrooms. Before they began, they were told that the test was organized by a renowned research institution and that the test results were not going to be disclosed to their teachers or parents. The tests were supervised by the experimenters to minimize the possibility of cheating by teachers and administrators. In each classroom, one teacher and one enumerator acted as exam proctors, and one or two additional enumerators walked around the test rooms as monitors. The tests were graded by computers.

The evaluation tests were designed to be similar to the baseline tests in format yet different in substance, taking into account that students had received an additional semester's worth of education.

The experiment focused on students in grades 3 through 6, as younger students would have had difficulties filling out the survey questions by themselves. Each grade-level typically has one to three classes. Within each school, the experimenters randomly selected several classes in grades 3, 4, 5 and 6, under the condition that each grade had one or two classes included in the study. In total, 82 classes were enrolled in the study. Every attending student in these classes participated in the tests and survey.

Each class in the study either hosted the pay-for-grades experiment (referred to as an *experiment class* or *treatment class*), or did not host an experiment at all (referred to as a *control class*). The experimenters then randomly divided these 82 classes into control and treatment groups, with at least one class per grade per school in the treatment group. To make sure that the control and treatment classes were truly similar to each other, the experimenters tested the between-group differences of the baseline test scores and 13 survey question responses using *t*-tests. If there was one *t*-test result that was statistically significant at the 90 % level (more conservative than 95 % in this context), the experimenters would

redivide the 82 classes into two groups randomly, and redo the *t*-tests until no *t*-test results were statistically significant at the 90 % level. This procedure guarantees that the treatment and control groups were balanced (i.e., similar to each other) at least in terms of the baseline test scores and the 13 survey responses.

Not everyone in the treatment classes received the treatment (i.e., the pay-for-grades program). Each experiment class had approximately ten treated students randomly selected from the group of the lowest performing twenty students based upon their baseline test scores.

The pay-for-grades program offered to the treated students resembled a merit-based competitive scholarship program. The experimenters promised to pay RMB 100 (approximately USD 13, or between one-third and one-quarter of a semester's tuition) to the treated students with the greatest increase in test scores between the baseline test and the evaluation test in the pay-for-grades experiment classes. The second and third place runners-up were promised RMB 50 each. This was a within-class competition. Only three treated students who made the largest improvement were entitled to a scholarship. Students were also promised a public ceremony and honorary certificates for the winners.

Students in control classes only took the baseline and evaluation tests; no program was offered to them. After dropping some classes for which seating tables were not available, we obtained 42 pay-for-grades treatment classes and 28 control classes that could be used to study cheating. Our analytical sample was comprised of a treatment group of the treated lowest performing students in these pay-for-grades classes and the control group of the lowest performing twenty students in these control classes.

Constructing a Cheating Index

We used the method developed by Wollack (1997, 2004) to detect cheating in our multiple-choice exams based on its high statistical power (Romero, Riascos, & Jara, 2014).

To calculate the ω index defined by Wollack (1997), a researcher must first identify a pair of two students, and then designate one of them as a potential copier and the other as a source. The index is only defined for the copier. If the researcher switches the copier and source statuses, the index will change. The

index for the copier is then constructed by comparing the observed number of identical answers and the theoretical number of identical answers predicted by a nominal response model under the assumption of no cheating.

Our context was friendly to the application of this index. First, we knew the seating arrangements for all students in our analytical sample. In contrast, Behrman et al. (2015) studied the effects of three types of cash incentive treatments (incentives to student only, incentives to teachers only, and incentives to both) on high school student test scores in Mexico using a randomized control trial. They found that higher rates of student copying contributed to positive treatment effects on learning. However, they had no classroom seating data and had to make assumptions about cheating behaviors. We did not have to make non-falsifiable assumptions about who copied from whom in a given classroom when no seating information was available. Second, we focused on the lowest performing students who were probably more likely to be copiers, not only because their higher-performing benchmates afforded them opportunities to do so, but also because their benchmates presumably had little interest in copying from them.

For a given pair of students, we, following Wollack (1997, 2004), define the ω index for student j (a potential cheater/copier) as

$$\omega_j = \frac{\text{(number of identical answers)} - \text{mean}}{\text{standard error}}, \tag{1}$$

where the mean and standard error are the estimated population mean and standard error of the underlying distribution for the number of identical answers between these two students if there is no cheating (i.e., if all matches of answers rise out of randomness).

The definition of the ω index follows the spirit of the standard hypothesis testing procedure. Suppose the null hypothesis is that no cheating occurs. In other words, all matches of answers rise out of pure randomness. The ω index is simply the familiar test statistic that measures how unlikely we may observe the data if the null hypothesis is true. We may reject the null if the test statistic is far away from zero. Wollack (2004) called cases of $\omega > 1.65$ as displaying some evidence of cheating. The pair of students must come up with a large percentage of identical answers to make $\omega > 1.65$. We may observe $\omega > 1.65$ purely by chance approximately 5 % of the time (i.e., the type-I error rate). Or, put differently, if no cheating occurs for this pair of students, the chances of us

observing such a large number of identical answers is only 5 %. In our analysis, we define a binary variable called *Cheater*, with

$$Cheater_j = \begin{cases} 1 & \text{if } \omega_j > 1.65, \\ 0 & \text{otherwise} \end{cases}. \tag{2}$$

It is important to emphasize that all statistical algorithms designed to detect cheating, including ours, are only able to detect severe cheating. For example, we may be able to detect a student copying 20 out of 30 questions from his/her benchmate, but are unable to detect a student copying 1 out of 30 questions. Of course, we acknowledge that copying even one answer constitutes as cheating, however, it is simply not possible for a statistical algorithm to detect such a mild case. Our definition of cheating indicator underestimates the existence of cheating. In other words, when our statistical method denotes someone to be a cheater, this person most likely did cheat; when our statistical method denotes someone to be a non-cheater, this person may still have cheated, but their cheating was not severe enough to catch the attention of the statistical algorithm. Because all tests in our study include a math portion and a Chinese portion, we calculated the cheating index for these two parts separately.

When the test questions are too easy and many benchmates end up with a large number of identical and correct answers, the index may overestimate cheating incidents. This is unlikely a serious issue in our context. Our tests are difficult for the migrant school students, who have a tendency to fall far behind the national standard. The rate of correct answer in the end-line math test has a mean of 0.62 and a standard deviation of 0.15. In other words, an average student only correctly answer six out of ten questions. The under-performing students do even worse.

Regression Specifications

We first studied how test scores were affected by the pay-for-grades program and other variables using the following regression equation

$$\text{score}_i = \alpha_1 + \beta_1 Treatment_i + \mathrm{X}_i \gamma_1 + \grave{o}_i, \tag{3}$$

where *score* represents the standardized math or Chinese scores from the baseline and evaluation tests. *Treatment* was the treatment status variable (1 for treatment students, 0 for control students). The matrix X_i included the following control variables: *Cheater*, *Pre-test* (the average of standardized math and Chinese scores from the baseline, used only when the outcome variables are

test scores in the evaluation tests), *Male* (1 for boys and 0 for girls), *Benchmate pre-test* (baseline Pre-test of the benchmate), as well as the grade dummies for each grade. Grade 3 dummy is defined as 1 for grade 3 and 0 otherwise. Grades 4, 5, 6 are defined similarly. Grades 4, 5, 6 dummies are included in the regression (Grade 3 dummy was dropped to avoid multicollinearity). Standard errors throughout the paper were clustered at the class level. β_1 captured the treatment effect on test scores. We used the following Probit regression to study to what extent cheating behaviors were affected by the existence of pay-for-grades programs and other variables:

$$\Pr(Cheater_i = 1) = \alpha_2 + \beta_2 Treatment_i + X_i\gamma_2 + \grave{o}_i, \quad (4)$$

where the matrix X_i includes the following control variables: *Pre-test*, *Male*, *Benchmate pre-test*, as well as the grade dummies. β_2 captures the treatment effect on cheating; the focus of this paper.

Every student has four observations in our data (baseline math, baseline Chinese, evaluation math, evaluation Chinese). In total, we have 610 students and 1,220 observations for either the baseline or evaluation stage. We first ran the above regressions for math and Chinese subsamples separately, and then for the full sample, at the baseline or evaluation stage. In the full sample case, we also added the math exam dummy variable (1 for math and 0 for Chinese) to capture any potential systematic differences across these two tests. We did not use the average of math and Chinese scores as a dependent variable. In specifications which include these scores, the results are essentially the same.

The baseline survey contained more variables than could be potentially added to the above regressions. We did not choose to add more variables because the analytical sample size would be further reduced when we added more control variables that contained missing values. More importantly, for a well-balanced experiment study with a large sample size, adding more control variables would not systematically change the size of the estimated treatment effects.

Main Results

Data Summary

Table 1 shows summary statistics for our data, by control and treatment status. A simple *t*-test result is also reported for each variable. For all variables except

Male, there were no significant differences between the control and treatment groups in the baseline stage (part A). As expected, students in our sample had lower test scores compared to their benchmates (see Table 1).

Table 1 Summary Statistics of the Analytical Sample

	Difference (1)	Control (2)	Treatment (3)
A: Baseline Pre-test	0.03	–0.76	–0.79
	(0.06)		
Benchmate pre-test	–0.04	–0.07	–0.03
	(0.08)		
Cheater (math)	0.02	0.19	0.17
	(0.03)		
Cheater (Chinese)	–0.01	0.09	0.10
	(0.02)		
Male	–0.08**	0.52	0.60
	(0.04)		
Grade	0.07	4.45	4.39
	(0.09)		
B: Evaluation Post-test	–0.08	–0.53	–0.45
	(0.08)		
Cheater (math)	0.00	0.06	0.06
	(0.02)		
Cheater (Chinese)	0.01	0.12	0.10
	(0.03)		
N	610	335	275

Note. "Difference" refers to the difference between control and treatment groups. Standard errors of *t*-tests reported in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Overall cheating intensities differ by subject and time. Cheating on the math test showed more fluctuations. The most intensive cheating occurred for the math test in the baseline period, with 19.1 % of the control group and 16.7 % of the treatment group defined as cheaters. However, in the evaluation period, these two numbers dropped to 6.3 % and 5.8 %. Cheating was relatively stable for the Chinese test (approximately 10 % in all cases). We will discuss the observed variations of cheating incidents in the next section.

Determinants of Test Scores

We first applied Regression 3 to the baseline sample, using baseline math and Chinese scores as the outcome variable. The results are reported in Columns 1 (math test only), 2 (Chinese test only), and 3 (both tests) of Table 2. The estimated effects of the treatment status were all insignificant. An insignificant effect of treatment status meant that there were no significant differences between the baseline test scores of the control and treatment students. This shows that randomization was well implemented. Being a cheater raised the baseline math test scores by 0.23 standard deviations (or SD). The effect was significant at the 1 % level. Cheaters' benefits on their Chinese test scores were also positive and close to being significant. Being a cheater raised math and Chinese scores combined by 0.20 SD (significant at the 1 % level).

Table 2 Regression Estimations of Incentive Treatment Effects on Test Scores

	Baseline			Evaluation		
	Math	Chinese	Both	Math	Chinese	Both
	(1)	(2)	(3)	(4)	(5)	(6)
Cheater	0.23***	0.16	0.20***	0.21	0.36***	0.28***
	(0.08)	(0.10)	(0.06)	(0.18)	(0.13)	(0.10)
Treatment	–0.02	–0.02	–0.02	0.11	0.11	0.11
	(0.09)	(0.09)	(0.09)	(0.09)	(0.09)	(0.09)
Male	–0.09	–0.07	–0.08	–0.11	–0.11	–0.11
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
Benchmate pre-test	0.16***	0.15***	0.16***	–0.04	–0.05	–0.04
	(0.04)	(0.04)	(0.04)	(0.05)	(0.05)	(0.05)
Math Exam			–0.02**			0.01
			(0.01)			(0.01)
Pre-test				0.57***	0.56***	0.567***
				(0.06)	(0.06)	(0.06)
Grade Dummies	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.05	0.04	0.05	0.21	0.22	0.22
N	610	610	1,220	610	610	1,220

Note. Figures above parenthesis are estimated coefficients. Figures in parenthesis are standard errors clustered at the class level. * $p < 0.1$, **, $p < 0.05$, *** $p < 0.01$.

Applying the above regression to the evaluation tests (with evaluation math and Chinese scores as the outcome variable), we found that the treatment effects

were still not significant. In other words, the pay-for-grades program failed to generate a significant impact on student test scores. This result mirrors the finding of Li et al. (2014).

More interestingly, we found that being a cheater, all other things being equal, raised a student's math and Chinese test scores by 0.21 and 0.36 SD, respectively (Columns 4 and 5). This effect on the Chinese test was statistically significant at the 1 % level. The effect on the Chinese and math tests combined was 0.28 SD (statistically significant at the 1 % level, Column 6).

In summary, student test scores were not affected by the existence of the pay-for-grades program. Cheating appeared to have a significant and positive impact on test scores in both the baseline and the evaluation tests.

Determinants of Cheating

Applying Regression 4 to the baseline sample, we obtained our baseline results, which are reported in Columns 1 (math test only), 2 (Chinese test only), and 3 (both tests) of Table 3. The estimated effects of the treatment status were all non-significant and consistent with our randomization design. The evaluation results are reported in Columns 4 to 6. We find that the treatment effects were all small and statistically insignificant. This result shows that students in our sample did not respond to monetary incentives tied to test scores by cheating.

Table 3 Probit Regression Estimations of Incentive Treatment Effects on the Cheating Likelihood

	Baseline			Evaluation		
	Math	Chinese	Both	Math	Chinese	Both
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment (d)	–0.02	0.03	0.00	–0.01	–0.00	–0.01
	(0.03)	(0.02)	(0.02)	(0.02)	(0.04)	(0.02)
Male (d)	0.02	–0.08***	–0.04	0.00	–0.03	–0.01
	(0.03)	(0.03)	(0.02)	(0.02)	(0.04)	(0.02)
Pre-test	0.07***	0.03*	0.04***	0.01	0.05**	0.02**
	(0.02)	(0.01)	(0.01)	(0.01)	(0.02)	(0.01)

(*To be continued*)

(*Continued*)

	Baseline			Evaluation		
	Math	Chinese	Both	Math	Chinese	Both
	(1)	(2)	(3)	(4)	(5)	(6)
Benchmate pre-test	–0.09***	–0.04***	–0.07***	–0.02**	0.01	–0.00
	(0.02)	(0.01)	(0.01)	(0.01)	(0.02)	(0.00)
Math Exam (d)			0.08***			–0.04
			(0.02)			(0.02)
Grade Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R^2	0.07	0.10	0.08	0.01	0.19	0.11
N	610	610	1,220	610	430	1,220

Note. (d) for discrete change of dummy variable from 0 to 1. Figures above parenthesis are estimated coefficients. Figures in parenthesis are standard errors clustered at the class level. A total of 180 observations for grade 5 Chinese test were dropped in the Probit regression because the outcome variable has no variation (all 0). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Was it possible that we failed to reject the null hypothesis of no treatment effect on cheating by chance when the null is actually true (i.e., commit a type-II error)? We calculated the statistical power, $1-\beta$. Defining the test statistic for the treatment effect under the null hypothesis that the treatment effect is zero as $\hat{T} = \frac{\hat{\beta}_2 - 0}{\hat{\sigma}}$, we used a two-tailed Z test. Power is a function of α (the significance level) and g (the true treatment effect) for a given sample size. It is easy to show (details available upon request) that for α = 0.05, the power for Equation 4 (for math and Chinese combined) was very close to 1 for $g > 0.1$. For g = 0.05, the power was still high, 0.84. For g < 0.05, the treatment effect becomes too small to be policy relevant. In summary, our statistical tests had high power for any reasonable level of true treatment effect. It is unlikely that our failure to reject the null hypothesis of no treatment effect was driven by a type-II error.

Across both periods, the baseline test scores appeared to be positively associated with cheating. We should not take this as evidence that students with higher test scores were more likely to cheat. First, all students in our sample were lower-performing students by definition. We cannot extrapolate these findings to other types of students. Second and more importantly, the causality probably

goes the other way around, namely cheaters tended to receive higher scores, as we discussed previously.

In both periods, the benchmates' baseline test scores, holding other variables (including a student's own baseline test scores) fixed, tended to be negatively associated with cheating. This negative association was statistically significant at the 1 % level in the baseline stage for the math-only subsample, Chinese-only subsample, and the full sample. In the evaluation stage, the association was still negative and statistically significant at the 1 % level for math. In other words, the test score gap between the student pairs tended to be negatively associated with cheating by the lowest performing students. All things being equal, a low-performing student is less likely to cheat when his/her benchmate has higher test scores. One possible explanation lies in the underlying social relationships between the student pairs. Suppose that a lower-performing student A wants to copy answers from his benchmate B. Cheating is more likely to go well with the help of the benchmate B. If B refuses to help, or even makes an effort to hide his or her answers, cheating is less likely to occur despite the intention of A. If the social distance between A and B is positively correlated with their test score gap, this helps to explain why cheating is negatively associated with the test score gap.

The positive correlations between social distance and test score gaps are a well-known phenomenon in most educational systems. In the US, this is usually framed as a racial issue—under-performing black students are more likely to interact with each other than with their higher-performing white peers (e.g., Echenique & Fryer, 2007). However, the problem can exist independent of the racial issue. Carrell, Sacerdote, and West (2013) showed that cadets of similar test score levels tended to form distinct social groups in the United States Air Force Academy despite the existence of an official policy that intentionally assigned them into mixed-ability peer groups. In our context, benchmate pairs are probably the most intensive social groups in a classroom. There is strong evidence that benchmate pairs tend to have similar test score levels. If we regress the benchmate test scores on a student's own test scores, we find that in the baseline period, when a student's own test scores increased by one SD, his benchmate test scores typically increased by 0.14 SD (statistically significant at the 0.1 % level, details available upon request).

In the baseline stage, there was significantly more cheating on the math test

than on the Chinese test. The estimated coefficient for the math test dummy was 0.08 and it was statistically significant at the 1 % level (Column 3 of Table 3). The estimated coefficient for the evaluation stage was small and statistically insignificant. These findings are consistent with the well-known result in the literature that cheating is more likely to occur in the math and sciences (Schab, 1991).

Except for the baseline Chinese test, being male did not have a significant impact on cheating. There was no strong evidence that cheating either increased or decreased by grade (details available upon request).

Discussion

Why did not cash incentives for test scores generate more cheating in our experiment? We offer two possibilities. The first is the age of students in our study. How children respond to various forms of incentives is not well understood (Rice & Broome, 2004). What we do know is that children may sometimes find non-cash incentives more appealing. The use of tokens (but not pennies, which are usually less expensive for experimenters and more valuable for children) as an incentive device or reinforcement system for both ordinary children and children with intellectual disabilities is widespread in psychology (Matson & Boisjoli, 2009). Children sometimes find small gifts, such as pencils and heart-shaped erasers, desirable; their tastes can change dramatically when they age (Hauber, Rice, Howell, & Carmon, 1998; Rice, Howell, Hauber, & Carmon, 1999).

In China, the use of red flower stickers to reward young students is common. It is entirely possible that other types of incentives—though not cash incentives—may encourage primary school students to cheat more.

The motivations behind student cheating probably matters as well. A dominant motivation of cheating is the fear of failure (Syam, 2014; Van Yperen, Hamstra, & van der Klauw, 2011). Less educated parents are more likely to use physical punishment when they are not satisfied with their child's outcome (Oreopoulos & Salvanes, 2011). Migrant parents are usually poorly educated and are more likely to use negative parenting styles, including spanking to discipline their children, than non-migrant parents in China (Zhang, Eisenberg, Liang, Li, & Deng, in press). Younger students are more likely to be spanked in China (Tao et al., 2004).

Promising primary school students cash rewards for good test scores may not encourage more cheating among low-performing students; on the other hand, it is possible that threatening them with severe punishment (such as spanking) in the case of exam failure might lead to significantly more cheating.

With these underlying motivations for cheating in mind, one possible explanation of why there was less cheating in the evaluation test was because students realized after the baseline test that the experiment was very low-stakes: Test scores were not reported back to their parents and there was nothing to fear. It also helps to explain why there was more cheating in math than in Chinese even when a multiple-choice format was used in both tests. We know that there is usually more cheating in math/science than in other types of exams (Schab, 1991). At first look, it seems to be just as easy for a student to copy a multiple-choice answer in a math test as in a Chinese test. However, the motivations behind cheating are not necessarily the same. Students tend to be nervous about math regardless of their math skills. Students with math anxiety may adopt avoidance strategies in math classes, including self-handicapping, disruptive behavior, and cheating (Hellum-Alexander, 2010). There is evidence that math anxiety is a more serious problem among migrant children in China (Ye, Wei, & Liang, 2012). Somewhat adding to the math anxiety is that the math questions in our study were slightly more difficult than the Chinese questions. In both baseline and evaluation tests, the average math correct rate is about 72 %, whereas the average Chinese correct rate is about 75 %. The difference in the correct rates is statistically significant at the conventional 95 % level.

The last possibility that may explain the lower occurrences of cheating is the design of our pay-for-grades program. We specifically designed the program to only award the three most-improved students in each classroom. In other words, students needed to compete with each other tournament-style, contrasting many other similar programs where all students will be rewarded once they reach a performance target (Slavin, 2010). Despite the advantage in research design discussed in Li et al. (2014), once students realized that not everyone would be offered a reward, the motive for cheating was likely to diminish.

Readers should be reminded that because our experiment focused on a particular grade range (i.e., grades 3–6), in theory the findings will be difficult to extrapolate beyond such population. It is possible that at middle school level, higher academic demands from the school and parents may ramp up students'

motivations for better grades. But for our particular sample, migrant children typically start to leave migrant schools during that precise period which leads to huge attrition (Chen et al., 2017). Therefore, we did not expand our population for that reason. This is something future research could further investigate.

Conclusion and Policy Implications

We used a randomized control trial implemented in migrant primary schools in China to evaluate the causal impact of providing under-performing students a pay-for-grades program on academic cheating. Here, cheating was defined according to a standard statistical algorithm that detects an abnormally large number of similar responses between a student and his or her benchmate. Our analysis showed widespread cheating behavior in standardized tests, especially on math tests and between bench pairs with relatively convergent ability levels. This pattern of academic cheating did not differ by gender or by grade. Despite this, we conclude that the pay-for-grades program did not increase the likelihood of cheating.

The implications of our findings are twofold. First, educators in China need to pay attention to various forms of academic cheating, particularly in low-performing schools. Our results suggest that cheating is not a case of just a few isolated incidents, but a common occurrence in our sample. Second, using moderate incentives to encourage student learning did not lead to skewed incentives. Our findings suggest that the pay-for-grades program such as the one we studied, will not have unintended consequence on cheating. Yet, we would also like to point out that pay-for-grades is likely context-dependent. Our design, which focuses on young primary students coupled with moderate financial incentives and low-stakes tests might not necessarily translate into other contexts. A case-by-case evaluation is needed before scaling up to other grade levels with different incentive targets.

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